Quiz2

This is a preview of the draft version of the quiz

- Due No due date
- Points 10
- Questions 9
- Available Oct 4, 2024 at 3:05pm - Oct 4, 2024 at 4:30pm 1 hour and 25 minutes
- Time Limit 35 Minutes

Instructions

The quiz will be conducted **in person, in class**. Remote access is not permitted.

The quiz is **closed-note**.

The use of **online resources (**including search engines and generative AI) is **strictly prohibited.**

Only the Canvas quiz page should be open during the quiz. Opening any other tabs will be considered an act of academic dishonesty, and the student will be escorted out and reported.

Do not share the quiz with students who are absent, as they may need to take a make-up quiz.

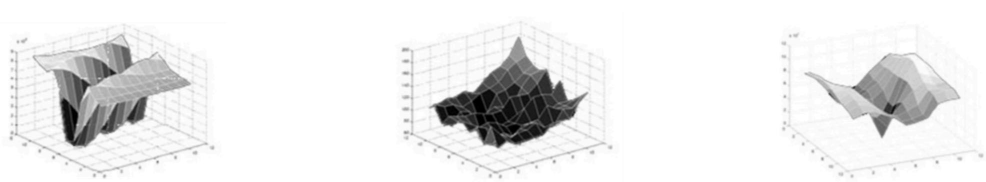
Preview

Score for this quiz: 0 out of 10
Submitted Oct 8 at 1:12pm
This attempt took less than 1 minute.
Unanswered

Question 1
0 / 1 pts

Correspond the three red crosses numbered 1,2,3 in the image to their autocorrelation function $E(u,v)$ shown as a surface plot:

$$E(u,v) = \sum_{x,y} w(x,y)\left[I(x+u, y+v) - I(x,y)\right]^2$$



(a) (b) (c)

1

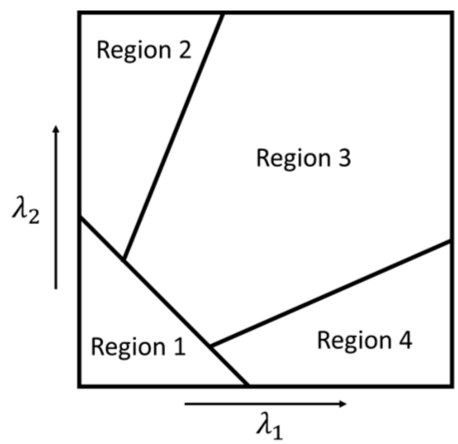
b
2

a
3

c
Unanswered

Question 2
0 / 1 pts

We want to classify keypoints in image I using eigenvalues λ_1, λ_2 of $M = \sum_{x,y} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$. Correspond each region to its correct characteristic.



Region 1

Flat
Region 2

Edge
Region 3

Corner
Region 4

Edge
Unanswered

Question 3
0 / 1 pts

In the image point classification problem described above, the "cornerness" score is computed as $C = \lambda_1 \lambda_2 - \alpha(\lambda_1 + \lambda_2)^2$. Match each region shown in the previous problem to the correct cornerness value.

Region 1

|C| small
Region 2

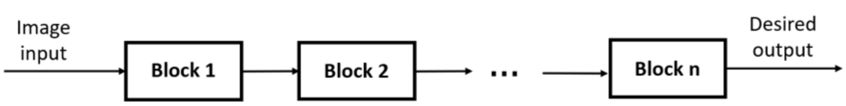
C < 0
Region 3

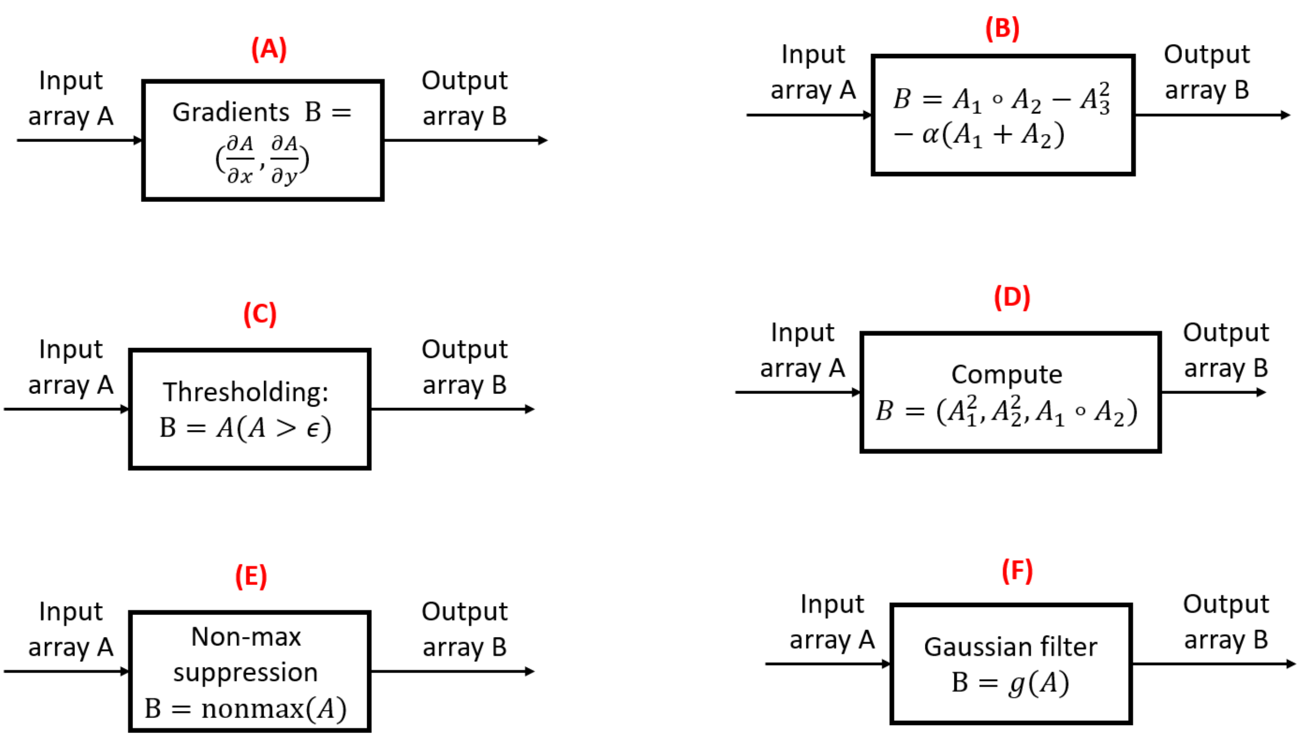
C > 0
Region 4

C < 0
Unanswered

Question 4
0 / 1 pts

Arrange the (unordered) blocks to construct a Harris corner detector algorithm:





Block 1

A

Block 2

D
Block 3

F
Block 4

B
Block 5

C
Block 6

E
Unanswered

Question 5
0 / 2 pts

Given 2 images with 3 keypoints in each image, we want to match the keypoints based on the Euclidean distance between their SURF descriptors, and, using the **Lowe's threshold of 0.7** to determine whether keypoints should be matched or not. Given the table of distances between all descriptors shown below, determine the correspondences between keypoints in image 1 to image 2, and vice versa.

$$\text{Image 1}\quad \begin{array}{c c} & \begin{array}{ccc} & \text{Image 2} & \\ q_1 & q_2 & q_3 \end{array} \\ \begin{array}{c} p_1 \\ p_2 \\ p_3 \end{array} & \begin{bmatrix} 10 & 1 & 100 \\ 10 & 20 & 100 \\ 30 & 100 & 90 \end{bmatrix} \end{array}$$

p1

q2
p2

q1
p3

q1
q1

none
q2

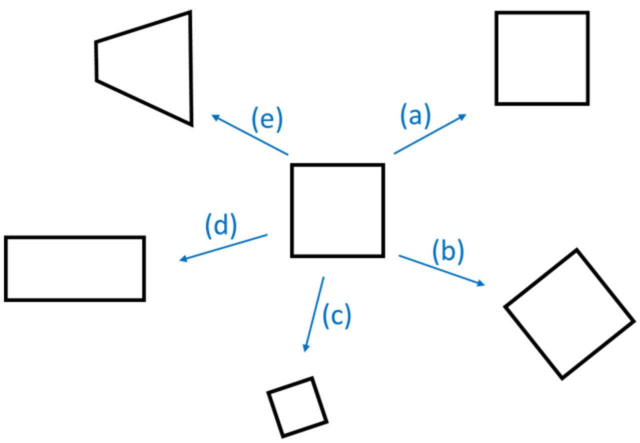
p1
q3

none
Other Incorrect Match Options:
- p3
- q3
- p2

Unanswered

Question 6
0 / 1 pts

Match 2D transformations based on how they transform (in the most general case) the shape of the square shown at the center:



a

Translation
b

Rigid body
c

Similarity
d

Affine
e

Projective

Unanswered

Question 7

0 / 1 pts

Recall the pinhole camera model

$$\mathbf{x} = \mathbf{P}\mathbf{X}$$

$$\underbrace{\begin{bmatrix} X \\ Y \\ Z \end{bmatrix}}_{\substack{\text{homogeneous} \\ \text{image coordinates} \\ 3 \times 1}} = \underbrace{\begin{bmatrix} p_1 & p_2 & p_3 & p_4 \\ p_5 & p_6 & p_7 & p_8 \\ p_9 & p_{10} & p_{11} & p_{12} \end{bmatrix}}_{\substack{\text{camera} \\ \text{matrix} \\ 3 \times 4}} \underbrace{\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}}_{\substack{\text{homogeneous} \\ \text{world coordinates} \\ 4 \times 1}}$$

Given the camera matrix $\mathbf{P} = \begin{bmatrix} 2 & 0 & 100 & 0 \\ 0 & 2 & 200 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$, and the 3D point $\mathbf{X} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$ what is the x-pixel coordinate of point $\mathbf{x}$ on the image? **Hint**: the x-pixel coordinate is *different* from the homogeneous image coordinate!

100 (with margin: 0)

Unanswered

Question 8

0 / 1 pts

Match each problem to the unknown(s) that it solves.

$$\underbrace{\mathbf{x}}_{\text{Unknown}} = \underbrace{\mathbf{K}[\mathbf{R} \quad \mathbf{t}]}_{\text{Known}}\underbrace{\mathbf{X}}_{\text{Known}}$$

(a)

$$\underbrace{\mathbf{x}}_{\text{Known}} = \underbrace{\mathbf{K}[\mathbf{R} \quad \mathbf{t}]}_{\text{Unknown}}\underbrace{\mathbf{X}}_{\text{Known}}$$

(b)

$$\underbrace{\mathbf{x}}_{\text{Known}} = \underbrace{\mathbf{K}[\mathbf{R} \quad \mathbf{t}]}_{\text{Known}}\underbrace{\mathbf{X}}_{\text{Unknown}}$$

(c)

$$\underbrace{\mathbf{x}}_{\text{Known}} = \underbrace{\mathbf{K}[\mathbf{R} \quad \mathbf{t}]}_{\text{Unknown}}\underbrace{\mathbf{X}}_{\text{Unknown}}$$

(d)

a

Camera simulation

b

Camera calibration

c

3D reconstruction
d

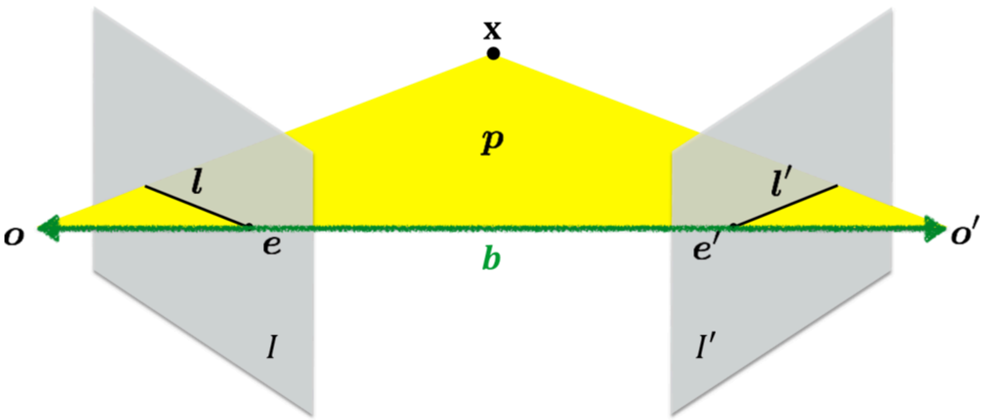
3D reconstruction
Other Incorrect Match Options:

- None

Unanswered

Question 9
0 / 1 pts

Correspond letters/notations to their names:



o, o'

Camera center
l, l'

Epipolar line
e, e'

Epipole
p

Epipolar plane
x

3D point
b

Baseline
I, I'

Image plane
Other Incorrect Match Options:

- Image center
- Fundamental matrix
- Essential matrix

Quiz Score: 0 out of 10